

# Mathematics

## CHAPTER II

*In which the strength of action according to figures is considered.*

AND since the pyramid, as has been said, is required for the action of nature, we must consider that the apex of the shorter pyramid acts more strongly, both because it is less distant from the agent, and because the rays conterminal around the apex of the shorter pyramid are in closer proximity, and proximity of rays and convergence act more strongly; and this is shown in the figure. For by the thirteenth proposition of the first book of the elements of Euclid, all the angles around a single point in a surface are equal to only four right angles, therefore the four angles at the apex of the shorter pyramid equal the remaining four at the apex of the longer pyramid. But by the twenty-first proposition of the same book, the angle in the apex of the shorter pyramid is greater than the angle in the apex of the longer pyramid, namely,  $a$  is greater than  $c$ ; and by the fifteenth proposition of the same book, angles placed opposite are equal, namely,  $a$  and  $b$ , likewise  $c$  and  $d$ ; therefore  $c$  and  $d$  taken together are less than  $a$  and  $b$  taken together; therefore since the four taken together equal the other four so taken by the thirteenth proposition, of necessity  $h$  and  $l$  are greater than  $f$  and  $e$ . Wherefore the rays containing  $e$  are in greater proximity than the rays containing  $h$ . And in the same way the rays containing  $f$  are nearer than the rays containing  $l$ , and so of an infinite number of rays conterminal in the shorter pyramid, and of necessity all of them are in closer proximity than the rays conterminal in the apex of the longer pyramid. But proximity of forces is the cause of stronger action. But, however, since the forward motion of perpendicular rays is the strongest, and every deflection toward perpendiculars is stronger than deflection away from them, the rays of the

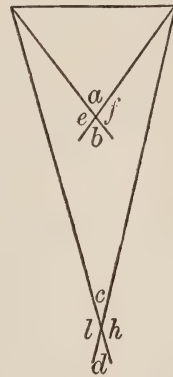


FIG. 7.

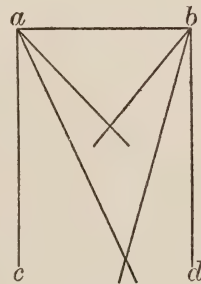


FIG. 8.